

## Builder Basics — The Trail Maker (Extension)

**Topic:** Place On Move · **Course:** Builder Basics · **Type:** Extension Activity · **Time:** about 45 minutes

✖ This is a bonus challenge, not a weekly lesson. Try it once you are comfortable placing blocks by hand and you are ready to let the agent place them for you.

### 1 · Predict 🧠

In the **Trail Maker** world, typing `run` in the chat starts the agent walking forward in a straight line. But first it flips a switch called **place on move**.

When **place on move** is **ON**, the agent drops a block on every single step it takes — it leaves a **trail** behind it as it walks. When the switch is **OFF**, the agent still walks the same steps, but it places nothing, so the ground stays empty and a **gap** appears.

Here is the exact order the program runs:

```
place on move ON
set block to purple concrete
move forward by 4
place on move OFF
move forward by 2
place on move ON
move forward by 4
```

**The switch starts ON and the agent moves forward 4. How many blocks does it drop in this first stretch?**

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**Then the switch turns OFF and the agent moves forward 2. How many blocks does it drop while the switch is OFF?**

**The switch turns ON again and the agent moves forward 4. Add up every block from the whole program. How many blocks in total? Show your adding.**

**Describe the finished trail. Where is the gap, and how wide is it? Draw or describe the shape.**

2 · Trace the Switch 

Walk through the program one line at a time. For each move command, write whether the switch is **ON** or **OFF**, how many steps the agent takes, how many blocks that stretch places, and the running total so far.

| Move command   | Switch (ON / OFF) | Steps | Blocks this stretch | Total so far |
|----------------|-------------------|-------|---------------------|--------------|
| move forward 4 |                   |       |                     |              |
| move forward 2 |                   |       |                     |              |
| move forward 4 |                   |       |                     |              |

Which move command placed NO blocks, and why?

The agent walked forward 10 steps in total but did not place 10 blocks. Explain why the number of blocks is smaller than the number of steps.

### 3 · Reason About Bugs

No code to rewrite here — just read what goes wrong and explain it in writing.

**Bug A — The switch never turns OFF.** Imagine the program leaves **place on move** ON the whole time and never flips it OFF in the middle. The agent still walks the same 10 steps forward.

**What does the trail look like now? What happened to the gap? Explain why.**

**Bug B — The switch never turns ON.** Imagine the program forgets the very first `place on move ON`, so the switch stays OFF from start to finish. The agent still walks all 10 steps.

**How many blocks get placed? What do you see on the ground when it finishes? Why?**

## 4 • Design Your Own Trail

Plan a trail of your own — in words and numbers, not in a code box.

**Choose your block, and decide your ON and OFF pattern. Fill in the blanks:**

My block is ...

Switch ON, move forward ... (places ... blocks)

Switch OFF, move forward ... (a gap ... wide)

Switch ON, move forward ... (places ... blocks)

**How many blocks does your pattern place in total? How many empty steps are in your gap?**

**Now the big idea. In Week 2 you placed each block by hand — one `place` command for every block. With `place on move` you write one `move` command and the agent places a whole line for you. In writing, explain WHY that is easier. To help you think: if you wanted a trail 50 blocks long, which way would you rather build it, and why?**

## 5 · Build It

Open the **Trail Maker** world and type `run` in the chat to watch the agent lay its trail. Then build your **own** version by changing the program — try a longer trail, a wider gap, more than one gap, or a different block.

Send a photo or video of your trail, then explain what you did. Use these starters — write 4 to 6 sentences total.

First, I ran `run` and saw the agent ...

When **place on move** was **ON**, the agent ...

When I turned it **OFF**, the agent ...

The change I made was ...

My trail has a **gap** because ...

Building this way was easier than placing by hand because ...

## 6 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) and teach someone how the Trail Maker works. You are narrating and explaining — not presenting to a class. Try to use these words: **place on move**, **switch**, **ON**, **OFF**, **trail**, **gap**.

1. Point to the trail and the gap, and say which part was built with the switch **ON**.
2. Explain what **place on move** does when it is **ON**.
3. Explain what happens to the ground when it is **OFF**, and why that made the **gap**.
4. Show the change you made and what it did to the trail.
5. Say one reason this is easier than placing every block by hand.

**Write what you will say in your video.** Plan it here first — you can read from it while filming.

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**Submit** 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.